AIM

To write a C program to add and multiply two matrices using pointers by accepting the number of rows and columns from the user.

PROCEDURE

1. Include the header file stdio.h.
2. Ask the user to enter the number of rows and columns of the matrices.
3. Read matrix elements using scanf().
4. Use pointers to perform matrix addition.
5. Use pointers to perform matrix multiplication.
6. Display the resultant matrices.
7. Compile and execute the program.

CODE

```
/*
Name of the File : pointer.c
Author           : Monish Kanna
Date             : 17-01-2026
Purpose          : To add and multiply two matrices using pointers
                   by getting rows and columns from the user.
*/
#include <stdio.h>
void addMatrix(int *m1, int *m2, int *result, int rows, int cols);
void multiplyMatrix(int *m1, int *m2, int *result, int rows1, int cols1, int cols2);
int main()
{
   int rows, cols;
   int matrix1[10][10], matrix2[10][10];
   int sum[10][10], product[10][10];
   int i, j;
   printf("Enter number of rows: ");
```

```
    scanf("%d", &rows);
    printf("Enter number of columns: ");
    scanf("%d", &cols);
    printf("\nEnter elements of first matrix:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix1[i][j]);}
   }
    printf("\nEnter elements of second matrix:\n");
    for (i = 0; i < rows; i++){
        for (j = 0; j < cols; j++){
            scanf("%d", &matrix2[i][j]);}
   }
    addMatrix((int *)matrix1, (int *)matrix2, (int *)sum, rows, cols);
    printf("\nMatrix Addition Result:\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++){
            printf("%d ", sum[i][j]);}
        printf("\n");
    }
    multiplyMatrix((int *)matrix1, (int *)matrix2, (int *)product,rows, cols, cols);
    printf("\nMatrix Multiplication Result:\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++){
            printf("%d ", product[i][j]);}
        printf("\n");
    }
    return 0;
}
```

```
void addMatrix(int *m1, int *m2, int *result, int rows, int cols)
{
    int i, j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            *(result + i * cols + j) =
                *(m1 + i * cols + j) + *(m2 + i * cols + j);} }
}
void multiplyMatrix(int *m1, int *m2, int *result,
                    int rows1, int cols1, int cols2)
{
    int i, j, k;
    for (i = 0; i < rows1; i++)
    {
        for (j = 0; j < cols2; j++)
        {
            *(result + i * cols2 + j) = 0;
            for (k = 0; k < cols1; k++)
            {
                *(result + i * cols2 + j) +=
                    (*(m1 + i * cols1 + k)) *
                    (*(m2 + k * cols2 + j));
            }
        }
    }
}
```

OUTPUT

```
PROBLEMS    TERMINAL    OUTPUT    PORTS

PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week2_practice5\" ; if ($?) { gcc pointer.c -o pointer } ; if ($?) { .\pointer }
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2

Enter elements of first matrix:
1 2 3 4

Enter elements of second matrix:
5 6 7 8

Matrix Addition Result:
6 8
10 12

Matrix Multiplication Result:
19 22
43 50
PS C:\AMI C PROGRAM\Monish_kanna_week2_practice5>
```

RESULT

Thus, the C program to add and multiply two matrices using pointers by accepting rows and columns from the user was successfully executed, and the correct output was obtained.